Lab 11 – Frequency Spectrum
EE252L/251L: Signals and Systems Lab

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Name: ____________________

Objectives

By the end of this lab, you will be able to get familiarize with the spectrum representation of signal using fast Fourier transform and spectrogram in MATLAB.

Instructions: Your code must be self-explanatory i.e. use proper comments. *Attach rubrics at the end of your report.*

Warning: This handout assumes that you have a fair idea of Continuous-time Fourier series and Discrete-time Fourier series. With this assumption it tries to build up to the ideas of different Fourier transforms in a simple way, but it is a tall order. So, this handout involves a greater reading component.

1 Spectrum

We know that a continuous-time periodic signal, $x(t)$ may be described as a weighted sum of complex exponentials, i.e.

$$x(t) = \sum_{m=-\infty}^{\infty} X_m e^{jk\omega_0 t}, \tag{1}$$

where $\omega_0 = 2\pi/T$ and T is period of the signal. This is called Fourier series synthesis and a_k are called Fourier series coefficients, determined using

$$X_m = \frac{1}{T}\int_0^T x(t)e^{-jk\omega_0 t}dt. \tag{2}$$

Looking at (1), we need the pairs $(m\omega_0, X_m)$ to construct any periodic signal $x(t)$. Each pair $(m\omega_0, X_m)$ indicates the size and relative phase of the complex exponential component contributing at frequency $m\omega_0$. This set, $\{\cdots, (-\omega_0, X_{-1}), (0, X_0), (\omega_0, X_1), (2\omega_0, X_2), \cdots\}$, of pairs is called the **spectrum** or **frequency-domain representation** of the signal. In contrast, the time-domain representation gives the values of the signal at every instant in time, whereas the frequency-domain representation should be seen as an equivalent and compact way to describe the signal as it contains all the information required to synthesize signal again with (1).

A graphical plot of the spectrum is much more revealing and can be represented by vertical lines at the appropriate frequencies. But the terms X_m are complex in general, so we need to construct two plots to represent the spectrum – a plot of magnitude $|X_m|$ against the frequencies f_m and a plot of phases $\angle X_m$ against the frequencies f_m. Notice that $f_m = m\omega_0$ is the frequency in Hertz. For example, the magnitude spectrum of a square wave is shown in Figure 1.

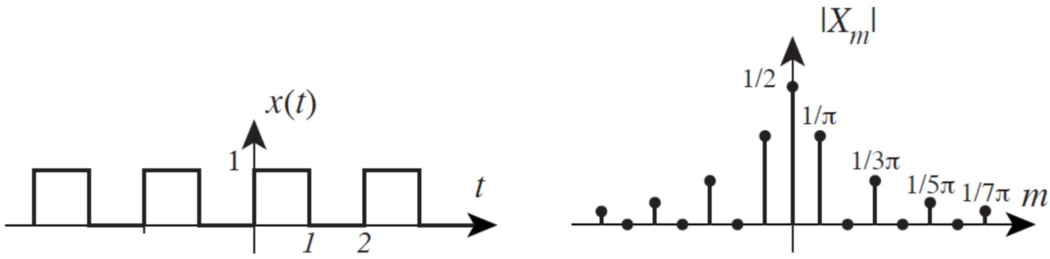

Figure 1: Magnitude spectrum of square wave

2 Discrete-Time Case

The idea of Fourier series extends to discrete-time periodic signals as well, and they can be represented as a sum of weighted complex exponentials too, i.e.

$$x[n] = \sum_{k=0}^{P-1} X_k e^{jk\omega_0 n}, \tag{3}$$

for any discrete-time signal $x[n]$ with period P, such that $\omega_0 = 2\pi/P$. Notice that we are only considering terms up to the index $P-1$ in the sum for discrete-time signal, in contrast to continuous-time case. This is because the terms repeat for $k \geq P$, e.g. $e^{jP\omega_0 n} = e^{j0\omega_0 n}$.

The coefficients, X_k, are given by

$$X_k = \frac{1}{P} \sum_{m=0}^{P-1} x[m] e^{-jmk\omega_0}. \tag{4}$$

The discrete-time Fourier series has the benefit that it is computable using a computer as opposed to continuous-time case where finding the coefficients requires the evaluation of an integral. Here, we need to simply compute a sum, and that too a finite (limited) one. Is there a MATLAB function? Yes, but let's relate this Fourier series to Discrete Fourier Transform first.

2.1 Discrete Fourier Transform

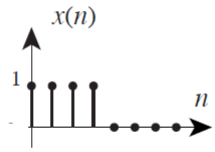

Figure 2: A finite discrete-time signal

What if our discrete-time signal is not periodic? If the discrete-time signal has finite length, (e.g. Figure 2), then we could construct a periodic signal from it by repeating the pattern (Figure 3) and then compute its Fourier series. This is called the **Discrete Fourier Transform**. Historically, the

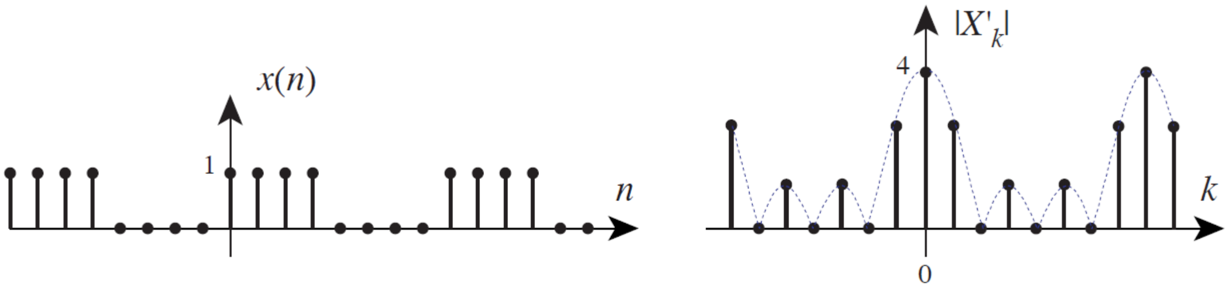

Figure 3: Discrete Fourier Transform

discrete Fourier transform was defined with a difference of scaling from the discrete-time Fourier series, and those historical expressions are still used today for consistency. The expressions are:

$$x[n] = \frac{1}{P}\sum_{k=0}^{P-1} X'_k e^{jk\omega_0 n} \tag{5}$$

$$X'_k = \sum_{m=0}^{P-1} x[m]e^{-jmk\omega_0}. \tag{6}$$

Compare these equations (6) with those of the DT Fourier series (3) and (4). Notice that

$$X'_k = PX_k,$$

i.e. DFT coefficients are simply scaled copies of DTFS coefficients. Note the periodicity of coefficients in Figure 3. Will this always be the case?

2.2 MATLAB functions

MATLAB provides functions `fft` and `ifft` to compute the DFT of a discrete-time signal and inverse DFT respectively. We can use the expression above to find DTFS coefficients too by scaling the ones obtained from `fft`. FFT stands for **Fast Fourier Transform**, which is a family of algorithms for computing the DFT efficiently. Straight computation of DFT from the formulas would take n^2 complex multiplications and $n(n-1)$ complex additions, where n is length of signal. Algorithms (e.g. Cooley-Tukey FFT) exploit symmetries of roots of unity to reduce calculations to $O(n\log_2 n)$. `fft` is fastest when n is power of 2.

Let's find the DFT of $x[n] = \cos[10\omega_0 n]$ using `fft`, where $\omega_0 = 2\pi/P$. Set $P = 64$.

```
P = 64;
w = 2*pi/P;
n = 0:P-1;
x = cos(10*w*n); %Finite-time signal
hx = fft(x); %Compute the DFT of x
stem(n,abs(hx)/P,'k','LineWidth',2); %Compute and plot magnitude of coefficients
ylabel('Magnitude of DFT');
xlabel('Frequency as multiples of \omega_0');
axis([0 P-1 0 inf]);
grid;
```

Does the plot of DFT agree with what you expected?

The plot contains an unexplained peak. These extra components are potentially the most confusing aspects of the DFT. But they are completely predictable from mathematical properties of DFT, specifically the DFT of real signal is conjugate symmetric. So,

$$|X'_k| = |X'_{-k}|.$$

Thus, if there is a peak at $10\omega_0$, we'll also get a peak at $-10\omega_0$. This is not visible as we're not displaying negative frequency axis, but since DFT is periodic the peak at $-10\omega_0$ appears again at $(-10 + 64)\omega_0$. We can display the DFT in symmetric form by shifting it using `fftshift`.

```
P = 64;
w = 2*pi/P;
n = 0:P-1;
x = cos(10*w*n); %Finite-time signal
hx = fft(x); %Compute the DFT of x
shx = fftshift(hx);
stem([-P/2:P/2-1],abs(shx)/P,'k','LineWidth',2);
ylabel('Magnitude of DFT');
xlabel('Frequency as multiples of \omega_0');
axis([-P/2 P/2-1 0 inf]);
grid;
```

Task 1-Magnitude Spectrum

Compute the magnitude spectrum (DFT) of $x[n] = \cos[10\omega_0 n] + 0.5\cos[14\omega_0 n]$, where $\omega_0 = 2\pi/P$ and $P = 64$.

(a) On Paper-(Refer Slides)

(b) On MATLAB

And also attach the scanned copy of your results in report.

2.3 Discrete-Time Fourier Transform (DTFT)

What if the discrete-time signal did not have finite length or what if we didn't want to repeat a finite length signal and instead wanted it to be zero everywhere else (Figure 4)? For cases

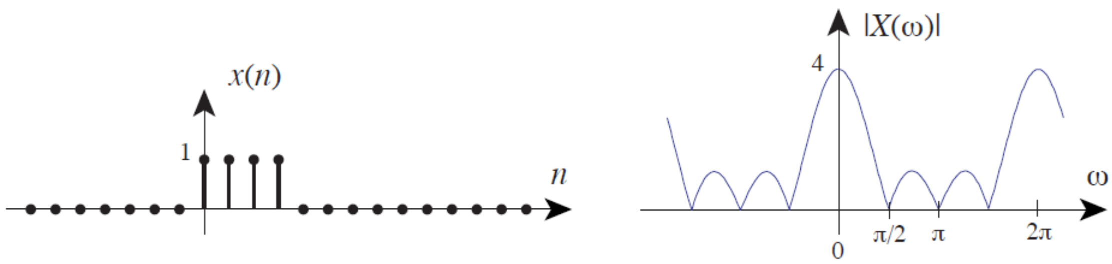

Figure 4: Discrete-time Fourier Transform

like this, we can use the **Discrete-time Fourier Transform**, which is different from the DFT. As evident from Figure 4, the DTFT is a continuous function. But notice that its shape seems similar to the DFT in Figure 3. This is not by accident and the DFT coefficients are samples of DTFT taken at integer multiples of ω_0. So we can use `fft` to find also the DTFT of signals by finding DFT and then fitting a continuous plot to it. Assuming that DFT samples are enough, we would be able to reasonably approximate the DTFT. How many samples are enough is a question for another day.

3 Continuous-time Fourier Transform

Analogous to discrete-time case, one can use the CTFT to find the spectrum of any continuous-time signal. You may have guessed that the expressions for this transform will involve integrals and so we have the same problem of computability. But the similarities between CTFT in Fig-

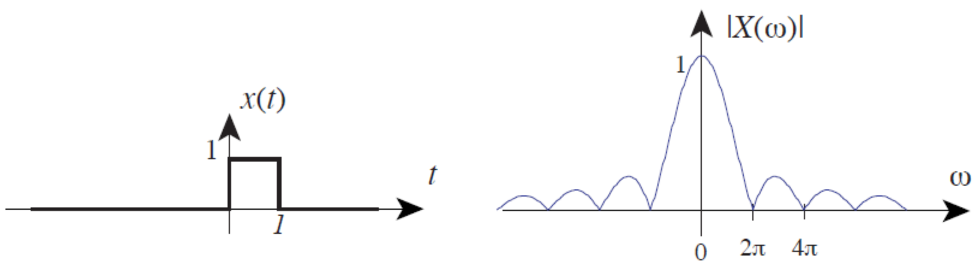

Figure 5: Continuous-time Fourier Transform

ure 5 and DTFT in Figure 4 gives us hope. Suppose you begin with a continuous-time signal $x(t)$ defined over a finite time period $[0, T]$ and then your obtain a discrete-time signal $x[n]$ by uniformly sampling the continuous-time signal $x(t)$. If N uniformly spaced samples are taken over the period T, then sampling frequency is $F_s = N/T$.

We can take the DFT of $x[n]$ and assuming that we have enough number of samples the CTFT can be approximated by this DFT. If the discrete-time frequency spectrum is on the range $[-T/2, T/2-1]\omega_0$ or $[-T/2, T/2-1]/T$ in Hz, then the continuous frequency range is obtained after multiplying this range by F_s.

Task 2

- Generate a sine wave signal `x` with frequency 400Hz sampled at 8kHz for $n = 1000$ samples.
- Take 1000 point FFT of `x` to produce `hx` and apply `fftshift` to `hx` so that zero frequency is in middle.
- Generate a vector `fHz` of frequencies in Hz corresponding to each point in `hx`.
- Plot a stem plot of the absolute value of the `fftshift`-ed `hx` vector against `fHz`. Is the result what you expected?
- Now do the same thing for frequencies 401 Hz, 402 Hz, and 440 Hz. What's special about 400 Hz and 440 Hz? *Hint: Plot the sine waveforms in time and think about period of each signal in discrete-time. Are we taking enough samples from one period?*
- Increase the number of points in FFT by using `fft(x,N)` where N is number of points. What effect does it have?
- Increase the number of samples of in discrete-time signal from 1000 to 4000 in the case of 402Hz. How does that affect the plot?

4 Spectrogram

Consider the signal $x(t) = \sin(2\pi 800 t^2)$. A signal of this kind is called a chirp signal. The instantaneous frequency of this waveform is:

$$\omega(t) = \frac{d}{dt} 2\pi 800 t^2 = 4\pi 800 t,$$

i.e. the frequency of the signal varies with time. Most real-world signals exhibit frequency changes over time. For instance, music may have a constant spectrum over some short time interval, but over the long term, the frequency content of a musical piece changes dramatically. In fact, this is what makes music pleasing to hear. A spectrum thus is unable to capture this time varying nature of music and we need some way to describe these time-frequency variations. One such time-frequency representation is the spectrogram. A spectrogram is found by estimating the frequency content in short sections (windows) of the signal. The magnitude of the spectrum over individual sections is plotted as intensity or color on a two-dimensional plot versus frequency and time.

In MATLAB, the function spectrogram can be used to compute the spectrogram. A common call to the function is `[s,f,t]=spectrogram(xx,1024,[],1024,fs,yaxis)`.The second argument is the window length which is varied to get different looking spectrograms and third argument is overlap between windows. In order to see what the spectrogram produces, run the following code:

```
fs=8000;
xx = cos(3000*pi*(0:1/fs:0.5));
spectrogram(xx,1024,[],1024,fs,'yaxis');
colorbar
```

The colors represent the magnitude in dB.

Task 3

1. Sample the chirp signal $x(t)$ as defined above at 8kHz and store 8000 samples of this signal in $x[n]$.

2. Find the FFT of this signal. How does it appear?

3. Find the spectrogram of this signal using a window length of 512 and 1024 FFT points. Use MATLAB help to get better understanding of function arguments. Does it agree with your understanding of the signal?

4. Find the spectrogram of `x(8000:-1:1)`. Does it make sense?

Assessment Rubric

Lab 11: Frequency Spectrum

Name:	**Student ID:**

Marks distribution:

	Tasks	**LR2 Code**	**LR5 Results**	**LR9 Report**	**AR4 Report Submission**	**AR7 Comments**
In-Lab	**Task 1**	/2	/4	/6	/12	
	Task 2	/8	/16	/8		/2
	Task 3	/4	/8	/8		/2
Total = 80		/14	/28	/22	/12	/4
CLO Mapped		**CLO2**	**CLO2**	**CLO2**	**CLO3**	**CLO3**

CLO – (LDL)	**Total Points**	**Points Obtained**
CLO2 – (Psy-4)	64	
CLO3 – (Aff-3)	16	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	*Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.
AR7	Report Content/Code comments	Most of the questions are not answered / figures are not labelled/ titles are not mentioned / units are not mentioned. No comments are present in the code.	Some of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Few comments are stated in the code.	Majority of the questions are answered, figures are labelled, titles are mentioned and units are mentioned. Comments are stated in the code.	All the questions are answered, figures are labelled, titles are mentioned and units are properly mentioned. Proper comments are stated in the code.